

Lecture

Developing Android Apps

Advanced GUI 2

Dr. Murad Mustafa Badarna

Material Design

Material Design הוא קו עיצוב שפותח על ידי Google שמטרתו ליצור חוויית משתמש עקבית, ברורה ויזואלית ואינטראקטיבית בכל הפלטפורמות Android, Web – iOS. גישה שמבוססת על עקרונות של תנועה, שכבות, עומק וצבע המדמים חומרים פיזיים.

מטרות העיצוב:

- **אחידות** בין אפליקציות וממשקים
- **נגישות** (accessibility) וקלות שימוש
- **תגובה אינטואיטיבית** לאינטראקציות משתמש (animations, transitions)

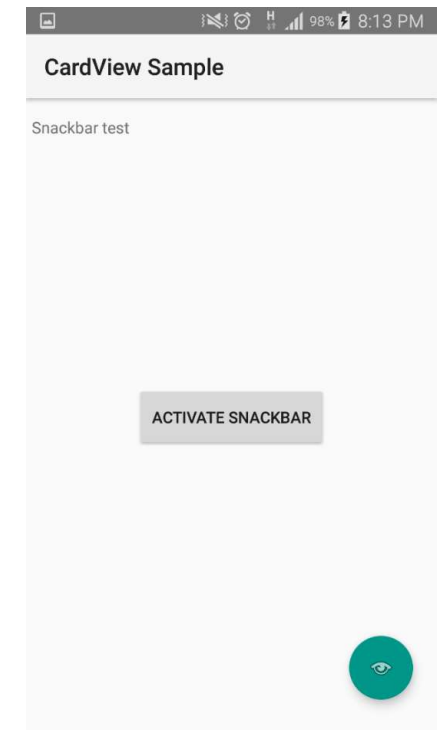
A new home for Material Design

Material is a metaphor, a system for uniting style, branding, interaction, and motion under a consistent set of principles. With Material we believe product teams can realize their greatest design potential.

- <https://design.google.com/resources/>
- <https://material.io/>

```
<android.support.design.widget.FloatingActionButton  
    android:id="@+id/fab"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:layout_alignParentBottom="true"  
    android:layout_alignParentRight="true"  
    android:layout_margin="16dp"  
    android:onClick="onShowListBtnClicked"  
    android:src="@android:drawable/ic_menu_view" />
```

```
public void onShowListBtnClicked(View view) {  
    Intent intent = new Intent(this, MyActivity.class);  
    startActivity(intent);  
    finish();  
}
```



Snackbar

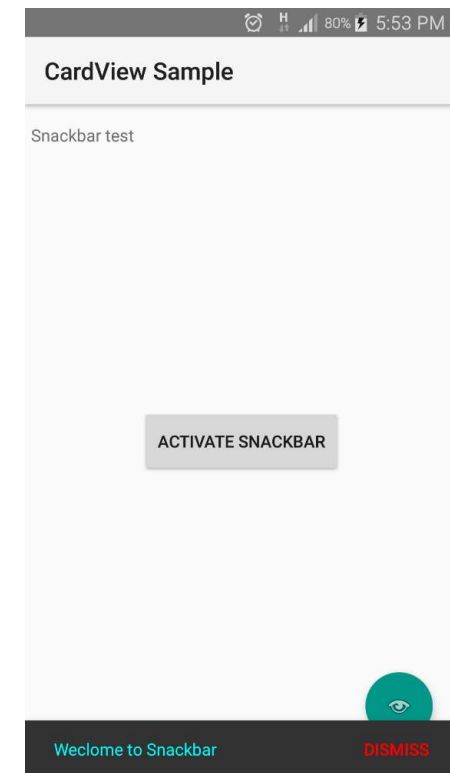
Must add to app gradle:
implementation
'com.google.android.material:material:1.9.0'

- **Snackbar** contain a single line of text directly related to the operation performed. They may contain a text action, but no icons.
- **Usage**
 - Only one snackbar may be displayed at a time. Each snackbar may contain a single action, neither of which may be “Dismiss” or “Cancel.”
- **Behavior**
 - Snackbars animate upwards from the bottom edge of the screen.

```
Snackbar bar = Snackbar.make(v, "Welcome to Snackbar", Snackbar.LENGTH_LONG)
    .setAction("Dismiss", new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            tv.setText("You pressed Dismiss!!");
        }
    });

bar.setActionTextColor(Color.RED);

TextView tv = (TextView) bar.getView().findViewById(android.support.design.R.id.snackbar_text);
tv.setTextColor(Color.YELLOW);
bar.show();
```



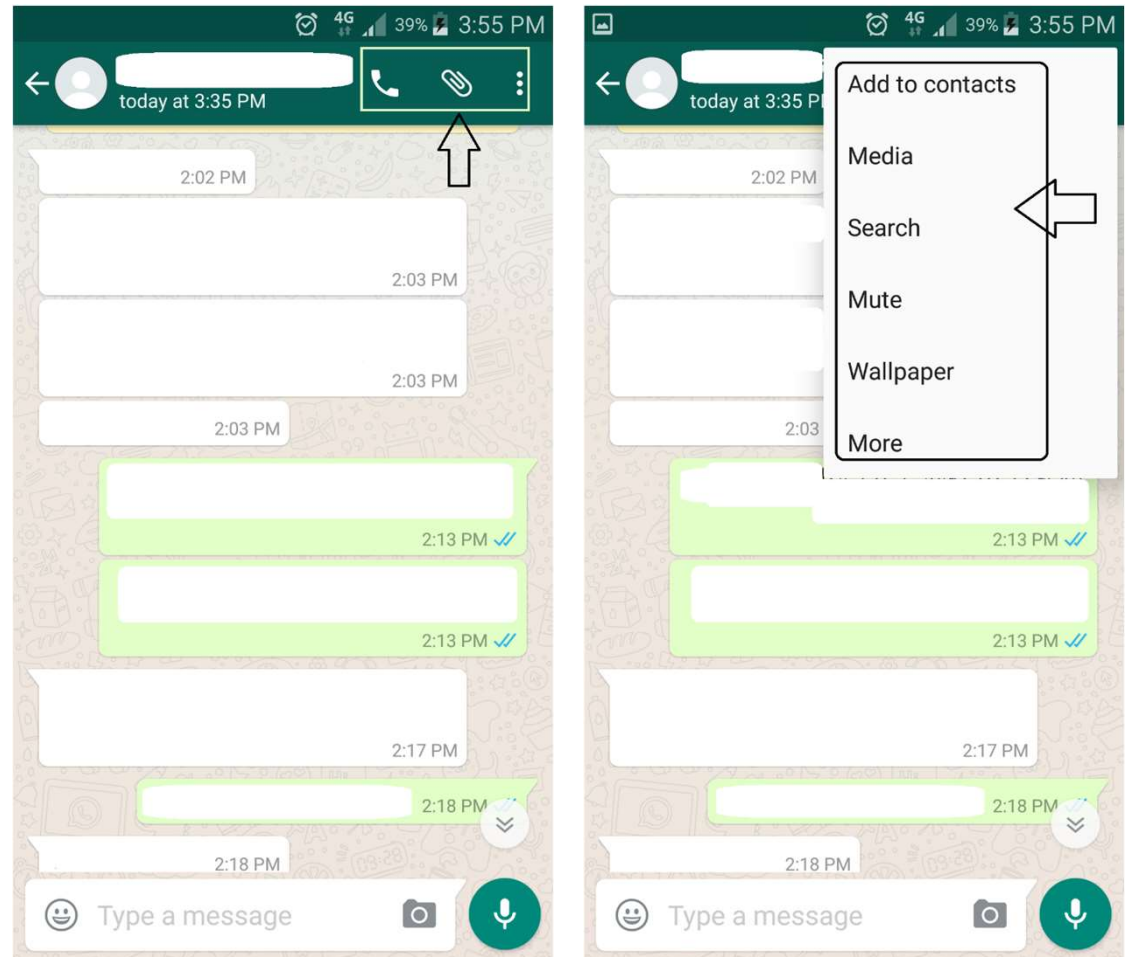
Import project called **MaterialApp2.zip** (download from our course website) into Android Studio

ToolBar

- The *tool bar* is a dedicated piece of real estate at the top of each screen that is generally persistent throughout the app.
- **It provides several key functions:**
- Makes important actions prominent and accessible in a predictable way (such as *New* or *Search*).
- Supports consistent **navigation** and view switching within apps.
- Reduces clutter by providing an action overflow for **rarely used actions**.
- Provides a dedicated space for giving your app an identity.

Lets look at the code of the demo app.

Import project called **ActionBarDemo.zip** (download from our course website) into Android Studio



Notification Status Bar

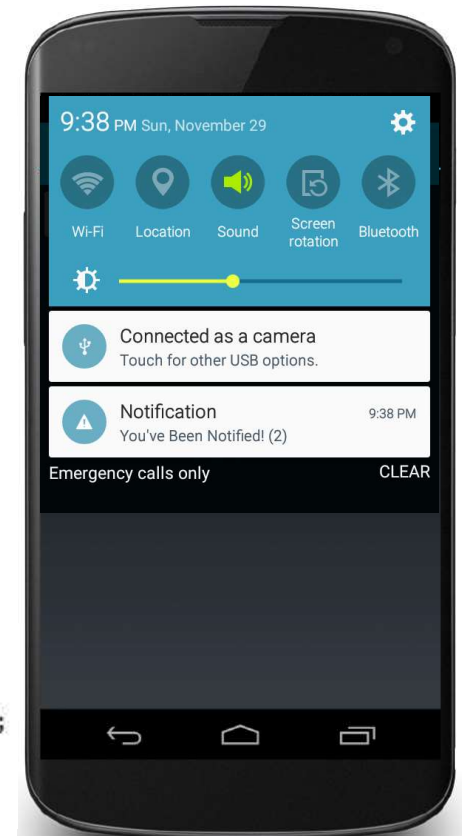
Work with
minSdkVersion 26

- Android provides the notification area for alerting users about events
- It also provides a notification drawer that users can pull down to see more detailed information about notifications

```
public class NotificationStatusBarActivity extends Activity {  
  
    // Notification ID to allow for future updates  
    private static final int MY_NOTIFICATION_ID = 1;  
  
    // Notification Count  
    private int mNotificationCount;  
  
    // Notification Action Elements  
    private Intent mNotificationIntent;  
    private PendingIntent mContentIntent;  
  
    @Override  
    public void onCreate(Bundle savedInstanceState) {  
        super.onCreate(savedInstanceState);  
  
        setContentView(R.layout.main);  
  
        mNotificationIntent = new Intent(getApplicationContext(), NotificationSubActivity.class);  
        mContentIntent = PendingIntent.getActivity(getApplicationContext(), 0, mNotificationIntent, Intent.FILL_IN_ACTION );  
  
        final Button button = (Button) findViewById(R.id.notify_button);  
        button.setOnClickListener(new OnClickListener() {  

```

A description of an Intent and target action to perform with it. This object can be handed to other applications so that they can perform the action you described on your behalf at a later time.

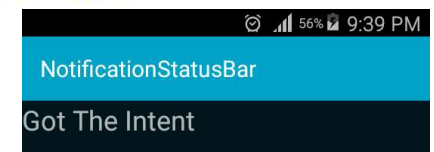
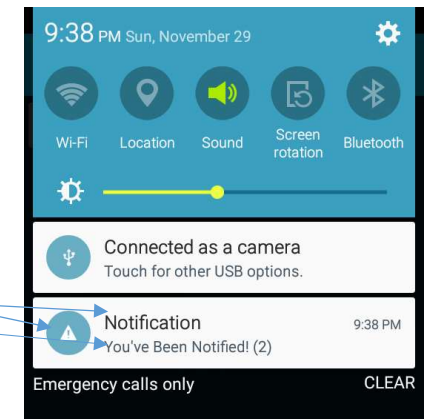
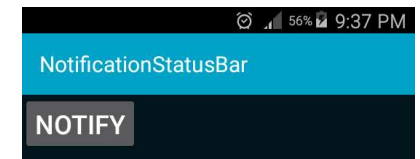


Import project called **NotificationStatusBar1.zip** (download from our course website) into Android Studio

Notification Status Bar

Must add to app gradle:
implementation
'androidx.core:core:1.10.1'

```
public void onClick(View v) {  
  
    // Define the Notification's expanded message and Intent:  
    // Notification Sound and Vibration on Arrival  
    Uri soundURI = RingtoneManager.getDefaultUri(RingtoneManager.TYPE_NOTIFICATION);  
    long[] mVibratePattern = { 0, 200, 200, 300 };  
  
    // Notification Text Elements  
  
    String contentType = "Notification";  
    String contentTypeText = "You've Been Notified!";  
  
    Notification.Builder notificationBuilder = new Notification.Builder(getApplicationContext());  
  
    notificationBuilder.setSmallIcon(android.R.drawable.stat_sys_warning);  
    notificationBuilder.setAutoCancel(true);  
    notificationBuilder.setContentTitle(contentTitle);  
    notificationBuilder.setContentText(contentText + " (" + ++mNotificationCount + ")");  
    notificationBuilder.setContentIntent(mContentIntent).setSound(soundURI);  
    notificationBuilder.setVibrate(mVibratePattern);  
  
    // Pass the Notification to the NotificationManager:  
    NotificationManager mNotificationManager = (NotificationManager) getSystemService(Context.NOTIFICATION_SERVICE);  
    mNotificationManager.notify(MY_NOTIFICATION_ID, notificationBuilder.build());  
}  
});
```



Import project called **NotificationStatusBar1.zip** (download from our course website) into Android Studio

ForegroundService – שירות בקדמה

- סוג של Service שנועד להריץ פעולות ארוכות טווח שכן חשובות למשתמש – לכן הן חייבות להיות מלוות בהתראה.

- **חייב להציג Notification קבועה** בזמן שהשירות פועל.

- **בעל עדיפות גבוהה יותר** – פחות סיכוי להיסגר על ידי מערכת ההפעלה.

- מתאים לפעולות כמו:

- ניווט. (GPS)

- הקלטת אודיו/וידאו.

- אפליקציית כושר או מוזיקה.

- העלאה/הורדה בקבצים ארוכים.

ForegroundService VS Service

תכונה	רגיל Service	ForegroundService
Notification דרישה ל-	לא	כן (חובה)
OS סיכוי להיסגר על ידי	גבוה	נמוך יחסית
נראות למשתמש	לא נראה	נראה (בהתראה)
שימושים מומלצים	משימות רקע קצרות	משימות רקע חשובות ומתמשכות